ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Paper 1 style (theory). Section A: objective questions (30) · Section B: magnetism and electromagnetic induction, transformers, cells and DC machines, resistance of materials, diodes, transistors, op-amps and capacitors, circuit prototyping and measuring instruments, logic gates, logic design and PLCs (70).

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The SI unit of magnetic flux is the …

A tesla. B weber. C henry. D ampere-turn.

1.1.2 The opposition that a magnetic circuit offers to magnetic flux is called …

A reluctance. B permeability. C resistance. D inductance.

1.1.3 A step-down transformer has …

A more turns on the secondary than on the primary.
B the same number of turns on both windings.
C fewer turns on the secondary than on the primary.
D no iron core.

1.1.4 A transformer can only work with …

A direct current.
B alternating current.
C the DC from a battery.
D static electricity.

1.1.5 Identical cells connected in parallel give …

A a higher voltage.
B a higher internal resistance.
C a smaller capacity.
D the same voltage with a larger current capacity.

1.1.6 A bipolar junction transistor used as a closed (ON) switch works in the …

A cut-off region. B active region. C saturation region. D breakdown region.

1.1.7 A Zener diode is mainly used for …

A amplification.
B voltage regulation.
C producing light.
D storing charge.

1.1.8 The input impedance of an ideal operational amplifier is …

A zero. B 1 kΩ. C 50 Ω. D infinite.

1.1.9 The instrument used to display the shape of a voltage waveform is the …

A oscilloscope. B ohmmeter. C insulation tester. D wattmeter.

1.1.10 The output of a 2-input NOR gate is HIGH (1) only when …

A both inputs are 1. B any input is 1.
C both inputs are 0. D the inputs are different.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Like magnetic poles attract each other.

1.2.2 For the same current, more turns on a coil give a larger magnetomotive force.

1.2.3 Laminating a transformer core reduces eddy-current losses.

1.2.4 A solar cell changes light energy directly into electrical energy.

1.2.5 The back emf of a DC motor increases as the motor speeds up.

1.2.6 Inductors connected in series have a total inductance smaller than the smallest inductor.

1.2.7 A voltmeter is connected in series with the load.

1.2.8 Veroboard is used for permanent, soldered circuits.

1.2.9 A multimeter set to measure resistance may safely be used on a live circuit.

1.2.10 The output of an AND gate is 1 when any one of its inputs is 1.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Ratio of magnetic flux density to magnetic field strength	A	capacitor
1.3.2	Device that changes the level of an AC voltage	B	PLC
1.3.3	Rotating part of a DC machine that carries the main winding	C	permeability
1.3.4	Split ring that reverses the current in the armature of a DC motor	D	Veroboard
1.3.5	Three-layer semiconductor device used to switch or amplify	E	armature
1.3.6	Diode that gives off light when it conducts	F	transformer
1.3.7	Component that stores energy in an electric field	G	LED
1.3.8	Board with holes and copper strips for soldered circuits	H	transducer
1.3.9	Device that changes a physical quantity into an electrical signal	I	commutator

1.3.10	Industrial controller that is programmed with ladder logic	J	bipolar junction transistor

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: MAGNETISM AND ELECTROMAGNETIC INDUCTION

2.1 Define the following terms:

2.1.1 Magnetomotive force (mmf) (1)

2.1.2 Reluctance (1)

2.2 A coil of 400 turns is wound on a ring-shaped core with a mean length of 0,25 m. The coil carries a current of 2,5 A. Calculate:

2.2.1 The magnetomotive force (2)

2.2.2 The magnetic field strength (H) (2)

2.3 State Faraday's law and Lenz's law of electromagnetic induction. (2)

2.4 The flux linking a coil of 200 turns falls from 0,05 Wb to 0,01 Wb in 0,2 s. Calculate the average emf induced in the coil. (3)

[11]

QUESTION 3: SUPPLY SYSTEMS, TRANSFORMERS, CELLS AND DC MACHINES

3.1 An ideal transformer has 1 150 turns on the primary winding and is connected to a 230 V supply. It must give 15 V on the secondary to power a robot charger. Calculate:

3.1.1 The number of turns on the secondary winding (2)

3.1.2 The turns ratio (1)

3.1.3 The secondary current when a 6 Ω load is connected (2)

3.1.4 The primary current (2)

3.2 Six cells, each with an emf of 1,5 V and an internal resistance of 0,2 Ω, are connected in series across an 8,8 Ω load. Calculate:

3.2.1 The total emf of the battery (1)

3.2.2 The total internal resistance (1)

3.2.3 The current in the load (2)

3.2.4 The terminal voltage of the battery (1)

3.3 Name TWO ways in which the field winding of a DC motor can be connected (excited) and give ONE typical application of each. (2)

[14]

QUESTION 4: ELECTRONIC COMPONENTS AND SEMICONDUCTORS

4.1 Calculate the resistance of a 50 m copper conductor with a cross-sectional area of 1,5 mm². The resistivity of copper is $1{,}72 \times 10^{-8}$ Ω·m. (3)

4.2 Explain what happens in a PN junction diode when it is forward biased and when it is reverse biased. (4)

4.3 Name the THREE terminals of a bipolar junction transistor and explain how an NPN transistor is used as a switch so that a microcontroller pin can switch on a 12 V relay. (4)

4.4 An inverting op-amp amplifier has a feedback resistor of 47 kΩ and an input resistor of 4,7 kΩ. The input voltage is 0,3 V. Calculate:

4.4.1 The voltage gain (2)

4.4.2 The output voltage (1)

4.5 Three capacitors of 10 µF, 20 µF and 20 µF are connected in series. Calculate the total capacitance. (2)

[16]

QUESTION 5: CIRCUIT PROTOTYPING, TOOLS AND MEASURING INSTRUMENTS

5.1 Compare a breadboard and a Veroboard. Give TWO differences. (4)

5.2 List, in the correct order, THREE steps you would follow to move a working breadboard circuit onto Veroboard. (3)

5.3 Explain how a digital multimeter is connected to measure the current through an LED, and give TWO safety precautions when measuring current. (4)

5.4 A sine wave on an oscilloscope screen measures 4 divisions from peak to peak, with the VOLTS/DIV knob on 2 V. One complete cycle covers 5 divisions with the TIME/DIV knob on 1 ms. Calculate the peak-to-peak voltage, the peak voltage, the period and the frequency. (4)

[15]

QUESTION 6: DIGITAL SYSTEMS, LOGIC DESIGN AND PLCs

6.1 Draw the truth tables for a 2-input NAND gate and a 2-input XOR gate. (4)

6.2 Use Boolean algebra to simplify $Y = A\cdot\bar{B} + A\cdot B + \bar{A}\cdot B$. Show ALL steps. (3)

6.3 A robot arm may only move (Y = 1) when the safety guard is closed (G = 1) AND either the start button is pressed (S = 1) OR automatic mode is switched on (M = 1).

6.3.1 Write the Boolean expression for Y. (2)

6.3.2 Name the logic gates, and how many of each, needed to build this circuit. (2)

6.4 Give TWO advantages of a PLC over a hard-wired relay control panel. (2)

6.5 State the function of the input module of a PLC. (1)

[14]

TOTAL SECTION B: [70]

TOTAL: 100

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END OF QUESTION PAPER